

# GDPR Data Anonymization Analysis

E-commerce Customer Data Pipeline  
Privacy-First Cloud Analytics

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April 2026

## Project at a Glance

~5,000

Customer Records  
Processed

8

PII Fields  
Anonymized

11

Anonymization  
Techniques

269

Free-text Notes  
Processed by Local AI

6

European  
Countries

100%

GDPR  
Compliance

# Executive Summary

01

## Scale & Scope

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~5,000 synthetic e-commerce records from 6 EU countries were processed through a full GDPR-compliant anonymization pipeline — from raw PII to cloud-ready, anonymized datasets.

02

## Techniques Applied

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11 anonymization techniques were combined: SHA-256 hashing, email & phone masking, date generalization, postal truncation, IP truncation, noise injection, field removal, and local AI (Ollama) for free-text notes.

03

## Value Preserved

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Despite full anonymization, 100% of analytical value is retained. 40+ queries across 4 purpose-limited views confirm that country, age range, product categories, customer segments, and behavioral patterns remain fully available for cloud analytics — with zero PII exposure.

# Anonymization Findings

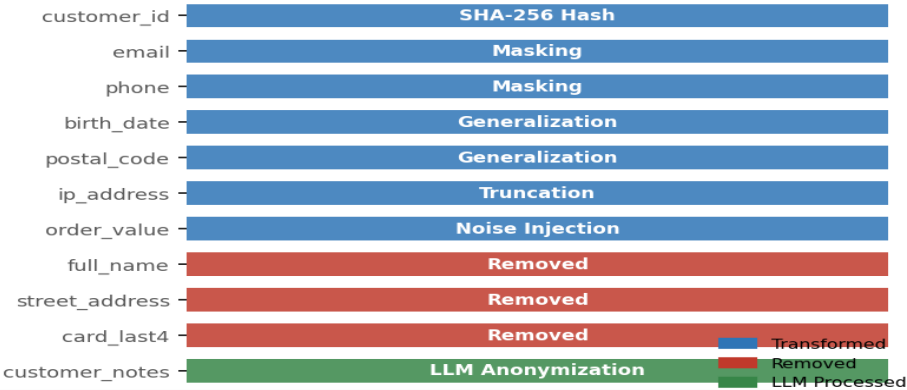
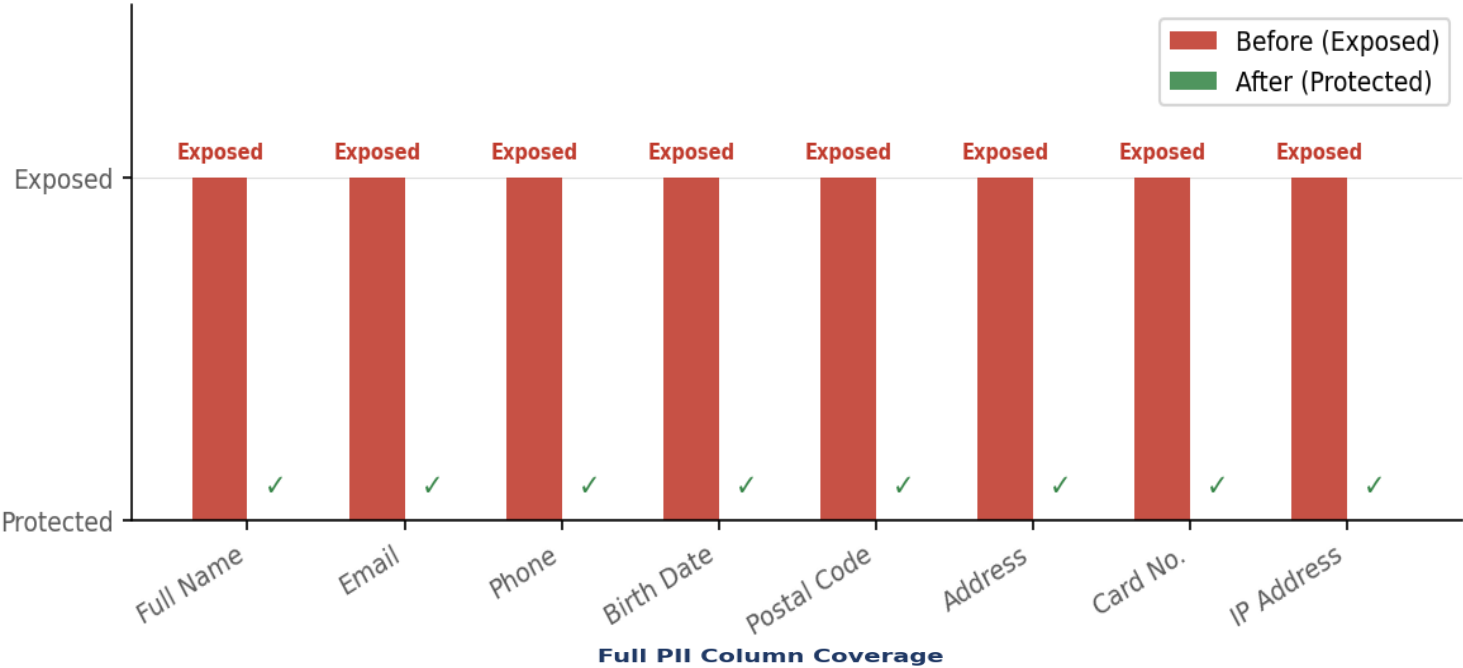
8 PII fields  
fully anonymized

3 fields  
permanently removed

269 notes  
processed by local AI

100%  
analytical value retained

8 PII Fields: Before vs After Anonymization



Before → After

| Field          | Before         | After           |
|----------------|----------------|-----------------|
| customer_id    | CUST-000764    | 919ed75...      |
| email          | user@email.com | us***@email.com |
| phone          | 0174-164776    | ***** 4776      |
| birth_date     | 1963-02-13     | 60-69           |
| postal_code    | N27 7NYE       | N2*****         |
| ip_address     | 161.198.44.35  | 161.198.0.0     |
| order_value    | 44.48 €        | 43.40 € ±noise  |
| full_name      | Lotte Blaak    | Removed         |
| street_address | Briandreef 461 | Removed         |

Free-text notes (269 records) processed with Ollama llama3.2:3b running locally — no customer data ever left the premises during AI processing.

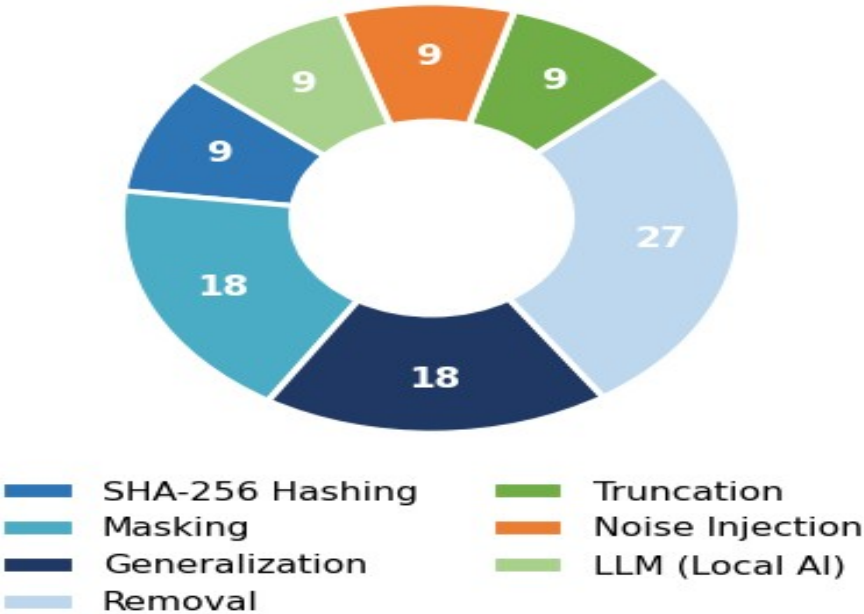
# Anonymization Strategy Applied

11 Techniques • 3 Categories • 1 Pipeline

|   | Technique             | Column         | Method                                        | Result      |
|---|-----------------------|----------------|-----------------------------------------------|-------------|
| 🔒 | SHA-256 Hashing       | customer_id    | Irreversible pseudonymization                 | Transformed |
| ✉ | Email Masking         | email          | Partial character replacement                 | Transformed |
| 📞 | Phone Masking         | phone          | Show last 4 digits only                       | Transformed |
| 📅 | Date Generalization   | birth_date     | Exact date → 10-year age range                | Transformed |
| 📍 | Postal Generalization | postal_code    | Full code → first 2 chars + *****             | Transformed |
| 🌐 | IP Truncation         | ip_address     | Last 2 octets zeroed out                      | Transformed |
| 🔢 | Noise Injection       | order_value    | Random ±offset added to EUR value             | Transformed |
| 🗑 | Field Removal         | full_name      | Deleted — no analytical value                 | Removed     |
| 🗑 | Field Removal         | street_address | Deleted — no analytical value                 | Removed     |
| 🗑 | Field Removal         | card_last4     | Deleted — no analytical value                 | Removed     |
| 🤖 | LLM Anonymization     | customer_notes | Ollama llama3.2:3b (local) → PII → [REDACTED] | AI          |

Pipeline order: Data Generation → MySQL Import → 8 SQL Anonymization Steps → k-Anonymity Validation → Analytical Views → LLM Text Anonymization → Cloud Export

Techniques Applied  
(11 total)



# Conclusions & Recommendations

## Privacy by Design Works



By applying 11 anonymization techniques at source, the NovaShop pipeline proves that GDPR compliance and cloud analytics are not mutually exclusive — you can have both without compromise.

## Local AI Is the Right Choice for Free Text



Processing 269 customer notes with a local LLM (Ollama) eliminated cloud PII exposure entirely. No sensitive data left the local environment — this approach should be standard for unstructured data.

## k-Anonymity Validation Closes the Loop



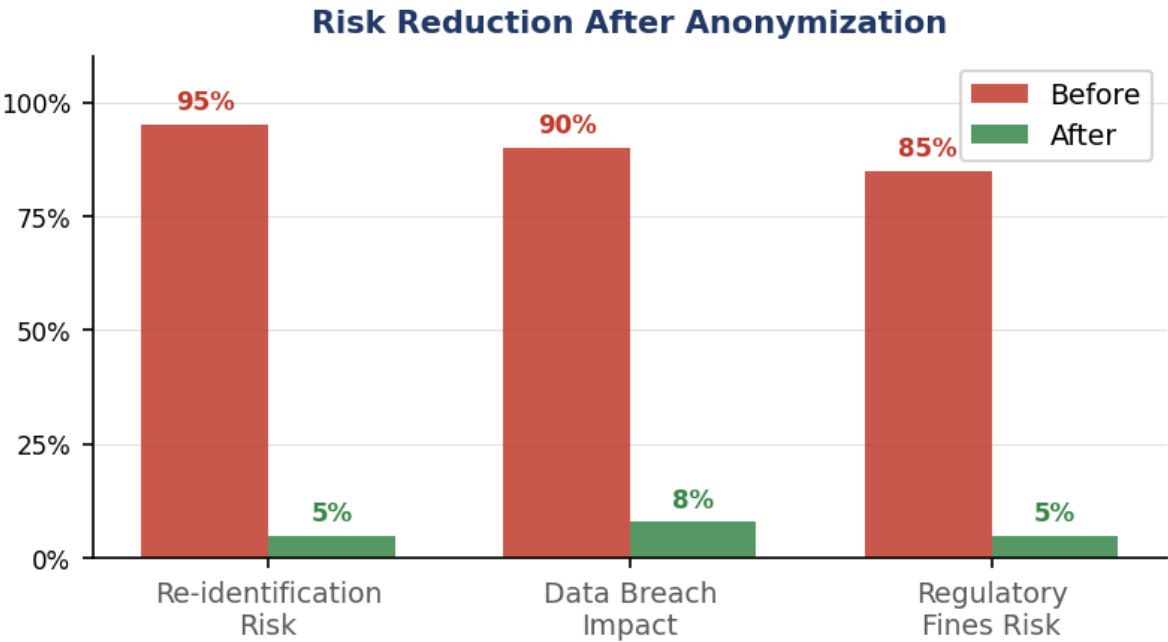
The pipeline validates k-anonymity after each transformation, ensuring no combination of quasi-identifiers can re-identify an individual. This is the difference between 'anonymous-looking' and 'legally anonymous'.

## Analytical Value Is Fully Preserved



Country, age range, product categories, device type, and customer segments all survive anonymization intact. 40+ analytical queries across 4 purpose-limited views validate this — enabling full BI and ML workloads on cloud platforms with zero compliance risk.

# GDPR & Security Benefits



## GDPR Articles Addressed

|         |                                                                                              |
|---------|----------------------------------------------------------------------------------------------|
| Art. 5  | <b>Data Minimisation</b><br><br>Only data necessary for analytics is retained.               |
| Art. 17 | <b>Right to Erasure</b><br><br>PII fields removed; individuals cannot be identified.         |
| Art. 25 | <b>Privacy by Design</b><br><br>Anonymization applied at the source, not as an afterthought. |
| Art. 32 | <b>Security of Processing</b><br><br>SHA-256, masking, and local AI prevent data exposure.   |
| Art. 89 | <b>Scientific Research</b><br><br>Anonymized data can be freely used for analytics & ML.     |

"Anonymize locally. Analyse globally."